

10. Write a Python Program to Implement Expectation & Maximization Algorithm

Aim

To implement the **Expectation-Maximization (EM) algorithm** for clustering data.

Software Required

- Python 3.x
- NumPy
- Scikit-learn
- Matplotlib

Procedure

1. Generate or load a dataset.
2. Select the number of clusters.
3. Apply the Gaussian Mixture Model.
4. The E-step estimates the probability of each point belonging to a cluster.
5. The M-step updates the parameters.
6. Repeat the process until convergence.
7. Display the cluster assignments.

Code

```
from sklearn.datasets import make_blobs
from sklearn.mixture import GaussianMixture

X, y = make_blobs(
    n_samples=100,
    centers=3,
    random_state=42
)

model = GaussianMixture(
    n_components=3,
    random_state=42
)

model.fit(X)
```

```

labels = model.predict(X)
print("Cluster Labels:")
print(labels)
print("\nCluster Means:")
print(model.means_)

```

Sample Output

```

... Cluster Labels:
[1 2 0 2 1 2 0 2 2 0 0 1 1 0 0 1 1 0 1 1 0 1 1 0 0 0 2 1 1 1 1 2 2 1 0 0 0
 0 2 2 1 0 2 0 0 2 1 1 1 2 2 2 0 1 1 1 0 0 2 0 1 2 1 2 1 1 2 1 2 2 2 1 1 0
 2 1 2 1 2 2 0 2 0 1 0 0 0 2 0 2 2 2 0 2 0 0 0 2 1 0]

Cluster Means:
[[-2.66780392  8.93576069]
 [-6.95170962 -6.67621669]
 [ 4.49951001  1.93892013]]

```

Result

Thus, the **Expectation-Maximization algorithm** was successfully implemented using a Gaussian Mixture Model.